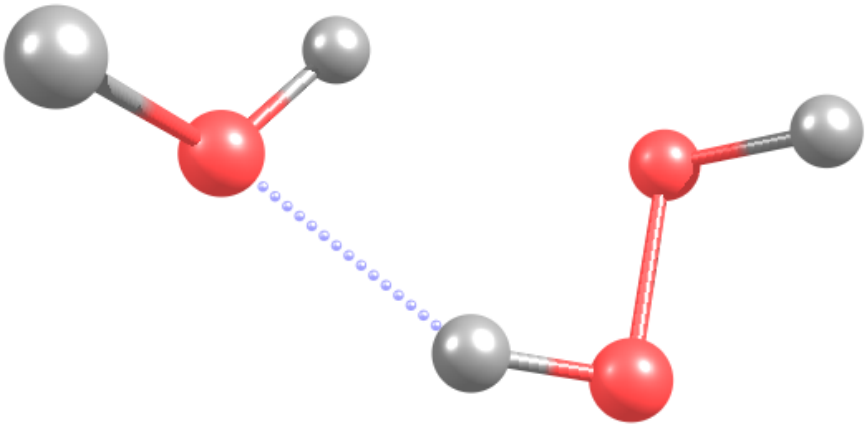
Laboratory Course Computational Physics, WS 24/25

Experiment: Monte Carlo Simulations of Molecular Systems



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1. Introduction

In the present experiment, the students will learn how large-scale chemical systems of more than thousands of atoms (e.g., enzymes, DNA, metal surfaces) are being modeled nowadays. The experiment will consist of a) writing a module for molecular mechanics simulation of water molecules, b) writing a module for Monte Carlo (MC) simulations in both a closed box and in periodic boundary conditions, c) writing a module to calculate radial distribution functions from an MC run, d) running simulations, and processing the results to obtain radial distribution functions at different temperatures. The tasks expect an implementation from scratch, with minimal use of sophisticated external libraries so that the students achieve a deeper understanding of the integral aspects of simulation codes.

The scheme of the experiment is as follows:

- First session: Discussion of force fields, Monte Carlo methods, radial distribution functions, and possible implementation problems. Starting the implementation.
- Second session: Finishing the implementation. Presentation of the code, discussion of possible shortcomings and lessons learnt.
- Protocol: A short summary of the implementation and obtained results are to be provided within two weeks after the second session.

The evaluation is based on the first session interview (20%), results from the two sessions (40%) and the protocol (40%).

Apart from the following information, it is expected that the students read the following chapters in *C. Cramer: Essentials of Computational Chemistry*: Chapters 2.2 and 2.3 (Molecular Mechanics) and Chapters 3.1–3.6 (sampling, molecular dynamics, Monte Carlo, radial distribution function). Understanding the covered topics is essential for the initial interview.

The programming language can be freely chosen by the students, with python being the preferred option.

2. Treating Large Systems: Molecular Mechanics and Sampling Problem

In modern chemical research, the field is constantly moving to the treatment of large-scale systems. Current quantum chemistry can provide very precise results on small systems of 1–6 atoms and reasonably precise results on systems with up to ~1000 atoms. However, this limits the calculations to gas phase systems (e.g. atmospheric chemistry) or invokes the need to treat the chemical environment in a simplistic way (e.g. polarizable continuum instead of proper solvent). However, many extremely important processes take place in solution or in bulk: Reactions in the car catalytic converter, organic synthesis, various degradation processes (e.g., corrosion) as well as virtually any biologically relevant process. For such systems, a set of approximate methods has to be developed.

Let us consider a chemically relevant system of an extensive size, say a hydrated DNA double helix. With the current techniques, it is impossible to treat such a system fully quantum mechanically, i.e., construct its wave function from first principles. Therefore, empirical approaches are needed. The first approximation would be to "ignore" the electrons, i.e., merge them into the nuclei, and treat atoms as billiard balls that interact through well-defined potentials, usually set so that experimental properties are reproduced. In the second step, one might even coarse-grain the problem further, e.g., ignoring the atomic structure and treating whole molecules or even larger aggregates in an approximated way. Here, we will deal only with the first approximation that is known in the literature as "molecular mechanics". One should also mention that machine learning might be used through modeling potentials trained on precise quantum chemical calculations. However, producing machine learning potentials is still often slower than molecular mechanics methods on larger systems, and therefore do not represent a considerable competitor to molecular mechanics at the moment.

The second problem when dealing with large systems is the extensive size of the conformational space. Even if one succeeded in performing a quantum mechanical calculation on a DNA system for a single conformation of the DNA, the statistical relevance of such a calculation would be virtually zero. In large systems, sampling is a crucial issue as one works with a high-dimensional space of ~3N nuclear coordinates in a system of N atoms. Sampling such a system at a finite temperature requires an efficient protocol. The two most popular

ones are molecular dynamics (MD) and Monte Carlo (MC). MD is a deterministic method that provides a way to sample the system in time, with the obvious advantage of obtaining process rates. However, it might get stuck in a part of the phase space that is irrelevant for the overall process and not cross into the more relevant part due to an energy barrier. Due to the molecular vibrations, MD time steps cannot be larger than a few femtoseconds, the length of MD trajectories is thus usually considerably limited. The MC ansatz, on the other hand, does not include time in its basic version, but provides a stochastic approach to retrieve the system properties through trying various steps and accepting or rejecting them based on the energy change. When the step types are chosen properly, one might circumvent the above-mentioned problem of getting stuck in a particular part of the phase space, as barrier crossing can be defined as a step in itself.

There is also a fundamental difference between MD and MC in implementation requirements. For MD at a finite temperature, a thermostat has to be implemented to provide "pushes" to the system to keep it on average at the desired temperature. At the same time, a careful choice of MD algorithm is needed to ensure proper integration of the equations of motion. In MC, the simulation temperature is set implicitly by the condition to accept or reject a proposed step, and no integration method is needed. Thus, MC implementation is much easier and was also chosen for the current experiment. Moreover, MC does not need to calculate forces on nuclei, saving significant resources in case forces are not available in an analytic form.

From an MC simulation of a molecular system, various types of information might be extracted. Here, we will limit ourselves to the radial distribution function (RDF). This function describes the distribution of neighbors in the vicinity of the molecule. Depending on the phase of the system, it might show very pronounced peaks typical for highly-symmetric solid phase, somewhat smeared-out structures in liquids, or be almost constant for the gas phase without pronounced space order (Figure 1). For example, the RDF is a popular tool in biology to analyze the interaction between large molecules and ions.

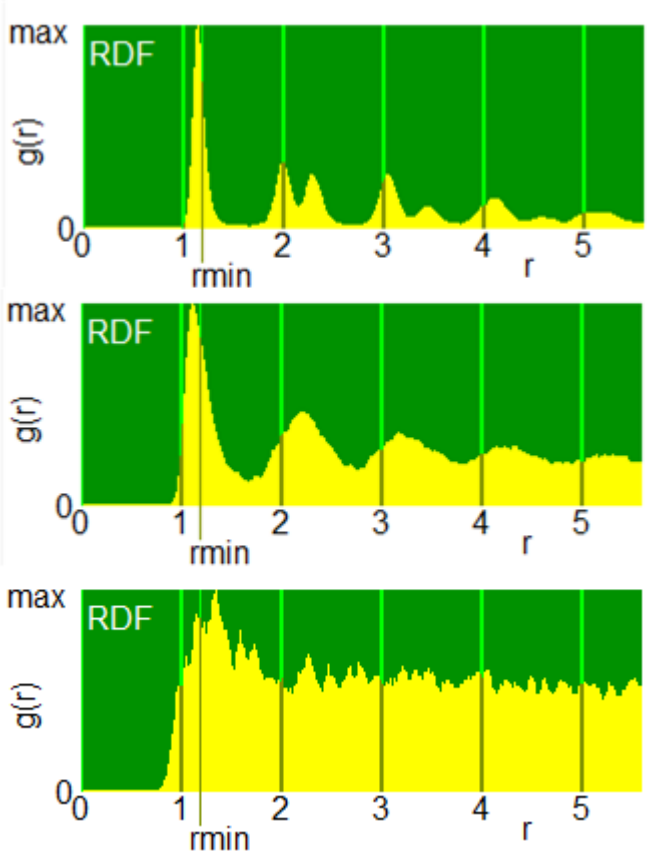


Figure 1 – Typical radial distribution function of solid (up), liquid (middle) and gas phase (down) systems as simulated with the Simolant software.

3. Molecular Mechanics in Action: Force Fields

Molecular mechanics is based on simplifying the interactions between electrons and nuclei, "ignoring" the electrons and postulating only several types of interactions between atoms instead of a full, quantum mechanical calculation. These interactions are usually:

a) *bond stretching*: When two atoms are covalently bound, the easiest way to describe their interaction as a harmonic potential in dependence on the interatomic distance *r*:

$$E_{\text{bond}} = \frac{1}{2}k(r - r_0)^2$$

b) *valence angle bending*: A usual form is again a harmonic potential with the angle α:

$$E_{\text{angle}} = \frac{1}{2}k(\alpha - \alpha_0)^2$$

c) *torsion*: For a torsion angle potential, usually a Fourier series is used.

d) *van der Waals interaction*: Interactions due to fluctuation of the electron distribution in atoms (attractive) as well as Pauli repulsion are usually described within a 12-6 potential (also known as Lennard-Jones potential):

$$E_{12-6} = \frac{A}{r^{12}} - \frac{B}{r^6}$$

e) *electrostatic interaction*: Even in neutral molecules, charge is usually not equally distributed and certain parts of the molecule are more negatively charged than others. The interaction between the charged atoms of nuclei that are not covalently bound can be described through a Coulomb potential:

$$E_{\text{Coulomb}} = \frac{1}{4\pi\varepsilon_0} \sum_{ij} \frac{q_i q_j}{r_{ij}}$$

The parameters in the equations above are fully empirical, set as to provide reasonable results for a given property of a system (i.e., they might differ even for water molecules, depending on which property one wants to simulate). The set of chosen equations and parameters to describe the interaction between atoms is called a *force field*, which is not be an intuitive term. There are many forces fields available in the literature and many new ones are created every year.

In the current experiment, we will describe a system of water molecules through the TIP3P model (Jorgensen et al., 1983). We will treat the water molecules as rigid, with the O–H distance of 0.9572 Å and the H–O–H angle of 104.52°. Therefore, only intermolecular interactions (cases d,e above) have to be accounted for. Parameters needed to describe the 12-6 potential are taken as $A = 582.0 \times 10^3$ kcal Å^{12}/mol and B = 595.0 kcal Å^6/mol, charges on atoms are defined as q(H) = 0.417 and q(O) = –0.834. These values were fitted to the experiment to reproduce the properties of the real water as much as such a simple system is able to.

Task #1: Write a module that calculates the energy of a system of *N* water molecules randomly placed in a box with and without periodic boundary conditions. In periodic boundary conditions, the energies should be calculated between each molecule and all its nearest neighbors. Consider only the 12-6 potential and Coulomb interaction as described above.

Task #2: Improve the module to calculate the energy *difference* between two systems of *N* water molecules upon change in the position of a single water molecule. This is exactly

the quantity that is later needed to decide if an MC step should be performed. Through this improvement, a lot of computational time can be saved as the majority of intermolecular interactions do not change upon the movement of a single water molecule.

4. Sampling the system: Monte Carlo

A Markov Chain Monte Carlo (in following called "Monte Carlo" for simplicity) simulation is based on the idea of trying different random steps and evaluating whether such a step should be taken or not based on a Boltzmann-type criterion. The sampling should then reproduce the relative probability of two states A and B at a finite temperature that is given by the Boltzmann factors:

$$\frac{P_\mathrm{B}}{P_\mathrm{A}} = \frac{\exp\left(-\frac{E_\mathrm{B}}{k_\mathrm{B}T}\right)}{\exp\left(-\frac{E_\mathrm{A}}{k_\mathrm{B}T}\right)} = \exp\left(-\frac{E_\mathrm{B}-E_\mathrm{A}}{k_\mathrm{B}T}\right) = \exp\left(-\frac{\Delta E}{k_\mathrm{B}T}\right)$$

A way to arrive at this distribution is to accept every step that leads to a state of lower energy, but to accept a step that increases the energy only with the probability of $\exp\left(-\frac{E_\mathrm{B}-E_\mathrm{A}}{k_\mathrm{B}T}\right)$. Such a scheme is shown in Figure 2. If the change is accepted, the system arrives at a new configuration. If the change is rejected, the system stays in its old configuration; however, this "staying in the same configuration" is counted as a step. When one performs an MC simulation with enough steps, one will thus arrive at a correct distribution at the given temperature.

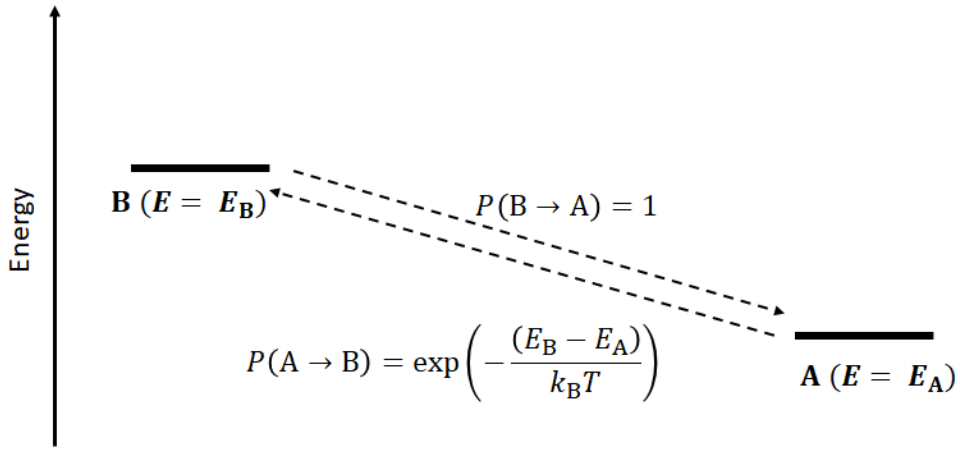


Figure 2 – Monte Carlo idea: Accept every step that lowers the energy, but allow the system to go up in energy only with the Boltzmann factor.

The "art of Monte Carlo" is to pick reasonable trial steps and reasonable trial step sizes. These should be picked to sample the system as efficiently as possible. Note that when a change is rejected, one has to evaluate the energy of a configuration that is subsequently not used. Generally, one does not want to accept too many changes as this means that the simulation is most probably doing too small steps and thus does not sample the possible configurations efficiently. On the other hand, a low number of accepted steps lowers the efficiency as well as the energy of many high-lying structures is calculated and subsequently discarded. Usually, one aims for an acceptance ratio of ~30%.

Task #3: Implement a Monte Carlo scheme at a given temperature for a box without periodic boundary conditions (i.e., box out of which the water molecules cannot escape). The box should be half-filled with water molecules (i.e., density of ~0.5 kg/dm^3) and the number of water molecules should be generated automatically depending on the box size. The initial position of water molecules is not critical as the MC algorithm will easily separate water molecules that are too close to each other. Consider two possible trial steps with a probability of 50%: translation of a water molecule in a random direction and rotation of a water molecule by a random angle. The user should be able to pick the range of the translation length and the rotation angle. Note that it must be impossible for a molecule to leave the box.

Task #4: Add periodic boundary conditions into the implementation above. Here, the system should be fully filled with water molecules as if one was simulating liquid water (density of ~1 kg/dm^3).

5. Is the simulation doing the right thing? Radial Distribution Function

A formal definition of the RDF between atoms A and B, g_{AB}, within a volume element is provided through a delta function:

$$g_{\mathrm{AB}}(r) = \frac{V}{N_{\mathrm{A}} N_{\mathrm{B}}} \langle \sum_{i=1}^{N_{\mathrm{A}}} \sum_{j=1}^{N_{\mathrm{B}}} \delta\left(r - r_{\mathrm{A}_i \mathrm{B}_j}\right) \rangle$$

We thus sum all occurrences of finding elements A and B within distance *r*, the number of atoms *N* and volume *V* are used for normalization purposes. An RDF provides us then with the probability *P* of finding atoms A and B within some distance *r* and range of Δ*r*:

$$P = \frac{4\pi r^2}{V} g_{\mathrm{AB}}(r)\Delta r$$

Probably the most intuitive definition of the RDF is as the ratio between local density at the distance *r*, $\rho(r)$, and the bulk density ρ:

$$g_{\mathrm{AB}}(r) = \frac{\langle \rho(r) \rangle}{\rho}$$

The usefulness of RDF should be apparent from Figure 1: It gives us information on the distribution of atoms in a simulation cell, being rather peaked for solids and liquids and almost constant for the gas phase.

Task #5: Create a module to calculate O–O, H–H, and O–H RDFs averaged over several distributions of water molecules that are provided as input.

6. Putting all together

Task #6: Perform a simulation of water molecules in a box without periodic boundary conditions. Pick a reasonable size of the system so that you are able to run ~100,000 steps. Run simulations at three different temperatures to obtain solid, liquid, and gas phase systems. Calculate the RDF averaged over a larger number of steps to check the convergence of the simulation (if the simulation is converged, the RDF should not change considerably).

Task #7: Simulate a system of water molecules in periodic boundary conditions and compare the simulation at two different temperatures to obtain solid and liquid water. Compare the RDFs in the two cases.

7. What has been achieved: Protocol

After a short introduction, the protocol should include a link to your program on the university Gitlab (https://git.uibk.ac.at) with a reasonable amount of comments as well as a description of any major decision you had to take during the implementation. Results from

Tasks 6 and 7 should be commented from the physical perspective (what was expected vs. what was achieved).

Literature

C. Cramer: Essentials of Computational Chemistry, Chapters 2.2, 2.3, 3.1–3.6 and references therein

J. Kolafa: Simolant, program for molecular dynamics and Monte Carlo, https://old.vscht.cz/fch/software/simolant/index-en.html (accessed on 22.9.2024)

W. L. Jorgensen, J. Chandrasekhar, J. D. Madura, R. W. Impey, M. L. Klein: Comparison of simple potential functions for simulating liquid water. *J. Chem. Phys. 79,* 926–935 (1983). DOI: 10.1063/1.445869